

University *of* Victoria Department *of* Mechanical Engineering

QT1.H Fuel System – Test Report
DOC-008 Rev. A

April 21st, 2026

Prepared For:

MECH 400b: Capstone Design Project

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1.0 Test Overview

All prototype testing was conducted on-site at the University of Victoria's Centre for Aerospace Research (CfAR), under direct client oversight. Four tests were conducted to validate the fuel system against the established engineering requirements. Note that only requirements directly addressed by prototype testing are referenced in this section; a full evaluation of all engineering requirements is provided in the detailed design document. A summary of each test and its outcome is provided in the table below, with detailed descriptions and results provided in the following section.

Table 1: Overview of Conducted Tests

Test	Goal	Result	Requirement Addressed
Static Leak Test	Verify all sealing surfaces; remain leak-free under static conditions	Pass	15
Pressurized Leak Test	Simulate pressurized flight conditions to confirm no leakage	Pass	15
Full Capacity Validation	Confirm tank holds full fuel load without leakage or structural concern	Pass	1, 2, 15
CG Testing	Determine the longitudinal and lateral %MAC change across fuel levels	Pass within ±1% MAC longitudinal and ±0.5% MAC lateral	8

All four tests were completed successfully with no leaks detected and all measured values within the specified requirements. Detailed test procedures, measurements, and results for each test are documented in the following section.

2.0 Test Description & Results

2.1 Static Leak Test

The static leak test was conducted to verify that all sealing surfaces remain leak-free under static conditions. The tank was filled to full capacity with water and placed on paper to allow visual detection of any leakage over time. The tank was periodically inspected over a 48-hour period to monitor for leaks. The tank was weighed before and after the test using a kitchen scale to quantitatively confirm no water loss occurred.



Figure 1: Fuel Tank Before (Left) and After (Right) Static Leak Test

The tank mass before and after the test was 1184g and 1183g respectively. The 1g difference is within the precision tolerance of the kitchen scale used and is not attributed to leakage. No leaks were visually detected throughout the 48-hour period, and the tank showed no signs of leakage on the paper beneath. The static leak test was passed with no leaks detected.

2.2 Pressurized Leak Test

The pressurized leak test was conducted to simulate pressurized flight conditions and confirm that the main seal and fitting seals operate as intended under internal pressure. The tank was filled to full capacity with water, and pressurized air was applied to one fitting at a constant 25 psig, with the remaining two fittings plugged for the duration of the test. Leaks were monitored visually, as any water leakage would be immediately apparent under the applied pressure.

No leaks were detected at any sealing surface or fitting throughout the test. The tank maintained 25 psig of internal pressure without any visible signs of leakage or sealing issues. The pressurized leak test was passed with no leaks detected.

2.3 Full Capacity Validation

The full capacity validation was utilized to determine the useable internal volume of the tank (including foam) and confirm that no leakage occurred when the tank is completely full. The tank was weighed empty and then filled to full capacity with water and weighed again.

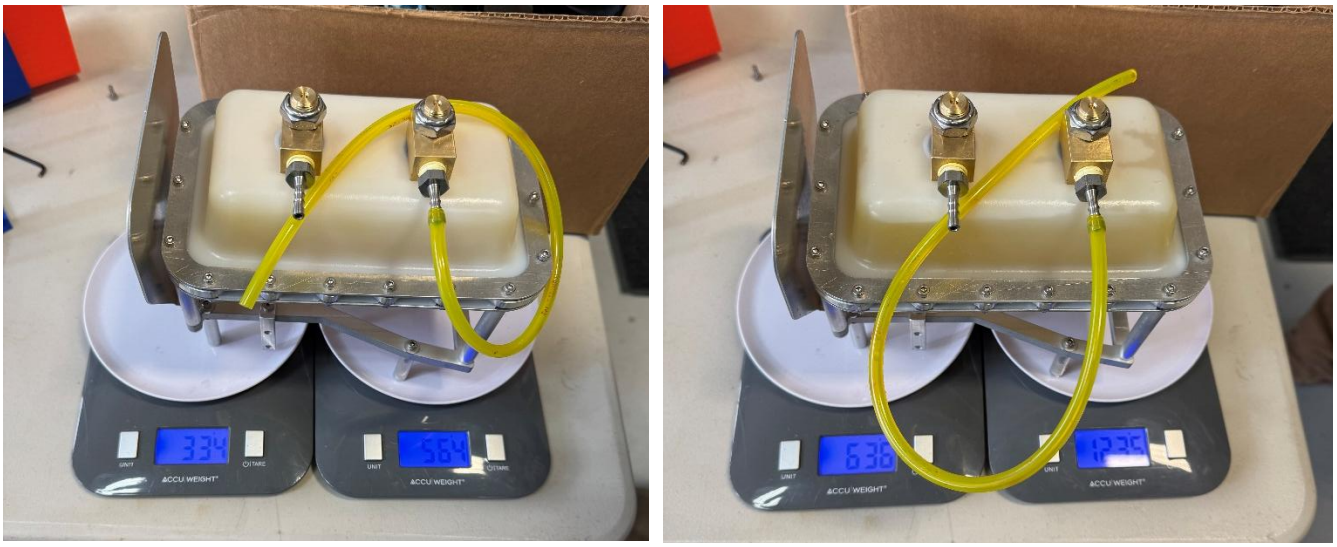


Figure 2: Empty (Left) and Full (Right) Fuel Tank Weight

The empty tank mass was 898g, and the full tank mass was 1871g, yielding a water mass of 972g. Using the density of water (1 g/mL), the internal tank volume can be determined.

$$\forall = \frac{m}{\rho} = \frac{972\ g}{1\ \frac{g}{mL}} = 972\ mL\ = 0.972\ L$$

With the internal tank volume determined, using the density of 94 octane gasoline (0.737 g/mL), the full fuel load can be determined.

$$m = \ \forall\rho = (972\ mL)\left(0.737\frac{g}{mL}\right) = 716.36\ g = 0.716\ kg$$

Therefore, approximately 716g worth of fuel can be utilized in the fuel tank. No leakage was observed during filling or while the tank was held at full capacity. The full capacity validation was passed successfully.

2.4 CG Testing and %MAC Calculations

Testing was conducted to ensure that the CG in the aircraft remains within the acceptable margin specified by CfAR. This was quantified using %MAC, calculated by studying the effect of changing CG. The primary requirements were to maintain a longitudinal CG shift of less than ±1% MAC and a lateral shift of less than ±0.5% MAC. For a comprehensive definition of %MAC, detailed CG calculations and interpretation of the final testing results, please refer to the detailed design documentation.

Horizontal CG testing was performed by weighing the fuel system at various increments from empty to full capacity, as seen in Figure 2 above, where the tank assembly was placed on calibrated scales. Using a fixed longitudinal spacing of 5.1 inches between the mounting posts, a 2D approximation was used to calculate the CG position relative to the datum (the front mounting posts). The recorded mass properties were tabulated to determine how the centroid shifted as a function of volume.

Table 2: CG Distance Properties

Left (g)	Right (g)	Mass (g)	CG (in)
334	550	884	3.125
334	564	898	3.155
363	623	986	3.174
369	649	1018	3.203
434	885	1319	3.371

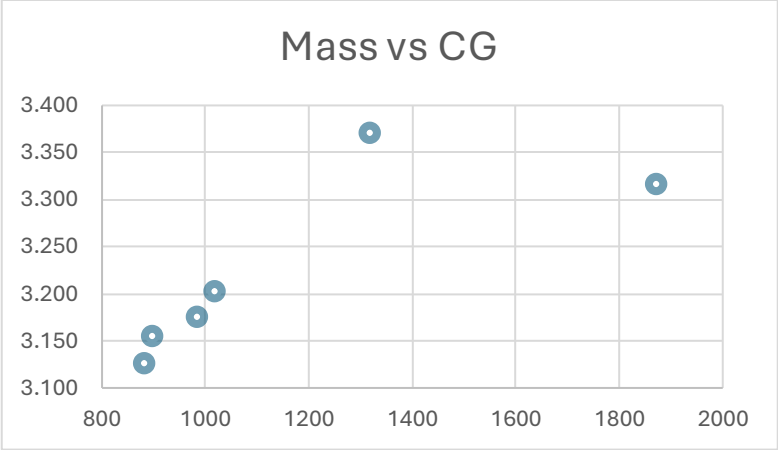


Figure 3: Mass vs CG

The results exhibited a specific trend correlating to the tank's internal geometry, which can be observed in Figure 3. During the initial filling of the triangular base, the CG shifted progressively away from the front mount. However, as the fluid level entered the upper rectangular section, the CG began to shift back toward the left post, effectively centering the total mass as the system reached capacity. To assess the worst-case scenario of %MAC from fuel slosh, the mass properties at 50% capacity were used, as directed by CfAR.

The analysis determined that the total longitudinal CG excursion under slosh conditions is ±0.1155% MAC, which is well within the ±1% limit. A lateral assessment also confirmed that variation remains below the ±0.5% MAC threshold. As a result, it was concluded that all CG shifts satisfied the original engineering requirements with significant margin.

3.0 Requirements Evaluation

The following section evaluates the fuel system against the engineering requirements directly addressed by the testing conducted. All requirements were confirmed as met. Note that only requirements directly addressed by prototype testing are referenced in this section. A full evaluation of all engineering requirements is provided in the detailed design documentation.

Table 3: Prototype Testing Requirements Evaluation

Requirement Number	Requirement	Measured Result	Status
1	< 5 kg, fuel included	1.614 kg total	Confirmed
2	< 2.5 kg, fuel included	1.614 kg total	Confirmed
8	CG < ±1% MAC longitudinal, < ±0.5% MAC lateral	±0.1155% MAC longitudinal, lateral confirmed within ±0.5% MAC	Confirmed
15	Compatible with 94 octane (40:1 gas/oil ratio)	No leaks detected under static and pressurized testing	Confirmed

The empty system mass was measured as 898g, with a calculated full fuel load of 716g, giving a total systems mass of 1614g or 1.614 kg. This is well within both the 5 kg and 2.5 kg targets.

At 50% fuel capacity, the maximum longitudinal CG change due to fuel sloshing was ±0.1155% MAC, well within the ±1% MAC requirement. Lateral CG variation was confirmed within ±0.5% MAC. The overall aircraft CG shift from full to residual fuel was 0.45% MAC.

Fuel compatibility was not directly tested with 94 octane fuel as CfAR was not ready to do so at the time of testing. All materials were selected for compatibility, and the tank produced leak-free performance under static and pressurized water testing. Fuel compatibility testing is recommended prior to flight.